

HE2003 Econometrics I

Tutorial 4 (Week 5) – Problems & Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

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Overview

- **Lecture coverage:** Lecture 4 — *Linear Regression with One Regressor 3*: further selection-bias examples (hospitals, wages, program evaluation, surveys, ratings); Least Squares Assumption 2 (i.i.d. sampling) and why simple random sampling matters; the measure of fit R^2 and its ESS/TSS decomposition; the Capital Asset Pricing Model (CAPM) and the “beta” of a stock as a running OLS application.
- **Questions:** 4 questions from Tutorial Question Set No. 4. Questions 1 and 3 have full official worked solutions on the Tutorial 4 slides (14 Sep 2026), reproduced and extended here. Questions 2 and 4 have *no* official solution – both “Question 2” / “Question 4” slides in the deck are blank – so both are solved here in full and flagged with
- **Notation:** as in lectures — u_i population error, $\hat{u}_i$ OLS residual; ESS explained sum of squares, SSR residual sum of squares, $TSS = ESS + SSR$; “Least Squares Assumption 1/2/3”; critical value 1.96 for a two-sided 5% test.

Question 1 (Selection bias examples; the measure of fit R^2)

Problem

- We discussed the selection bias/survivorship bias in class. Can you give some other examples of this bias?
- Explain the measure of fit R^2 . Illustrate the meaning of explained sum of squares (ESS) and total sum of squares (TSS) by using a diagram.

(Tutorial Question Set No. 4, Q1.)

Solution

Part (a): further examples of selection/survivorship bias

Method 1: the official example – investment gain and experience. Write the population regression as

$$\text{gain}_i = \beta_0 + \beta_1 \text{experience}_i + u_i, \quad i = 1, \dots, n,$$

where gain_i is investor i 's cumulative gain in the financial market, experience_i is i 's number of years of investment experience, and u_i collects everything else that drives gains, most importantly i 's *innate investing ability*. Least Squares Assumption 1 requires $E[u_i | \text{experience}_i] = 0$ for every level of experience. This fails here: investors with low ability tend to lose money and exit the market early, while investors with high ability are more likely to survive and keep accumulating years of experience. Consequently ability (part of u_i) is *positively* correlated with experience_i – only the survivors are observed at high experience levels, so the sample of “experienced” investors is not representative of what a randomly chosen investor would look like after that many years. Because $\text{Cov}(\text{experience}_i, u_i) > 0$, $\hat{\beta}_1$ is biased *upward*: part of the estimated “return to experience” is really a return to the unobserved, survivorship-selected ability of those who remained in the sample.

Beyond the slides: three further examples, in different fields. The official example is a template: (i) write down the regression, (ii) identify what is folded into u_i , (iii) explain why the sample that ends up being observed depends on u_i itself, (iv) sign the resulting bias.

- **Example 2 (finance): mutual-fund performance databases.**

- *Model*: $\text{return}_i = \beta_0 + \beta_1 \text{track_record}_i + u_i$, where return_i is a fund's reported average annual return and track_record_i is its number of years listed in a commercial database.
- *What is in u_i* : the fund manager's stock-picking skill and risk-taking.
- *Why Assumption 1 fails*: funds that perform badly are disproportionately liquidated, merged away, or dropped from commercial databases (“backfill” or survivorship bias); only funds that performed well enough to keep operating remain in the sample long enough to build up a long track record. So high track_record_i is systematically paired with high manager skill (high u_i): $\text{Cov}(\text{track_record}_i, u_i) > 0$.

- *Bias direction:* $\hat{\beta}_1$ (and the sample-average return more generally) is biased *upward* – this is precisely why headline mutual-fund industry returns computed from “funds that still exist today” tend to overstate what an investor could actually have expected to earn.

• **Example 3 (healthcare): hospital quality and patient outcomes.**

- *Model:* $\text{outcome}_i = \beta_0 + \beta_1 \text{hospital_quality}_i + u_i$, where outcome_i is a discharged patient’s measured health outcome (this generalises the Lecture 4 hospital-visit survey, where naively comparing “visited hospital” vs. “did not visit hospital” mean health status suggested hospitals make people sicker).
- *What is in u_i :* the patient’s underlying severity of illness on admission.
- *Why Assumption 1 fails:* patients who are too sick to survive treatment, or who are transferred to a different facility, are dropped from the outcome sample used to rank hospitals; which patients remain observable is correlated with u_i (severity), and if the sickest patients also sort into particular hospitals, $\text{hospital_quality}_i$ is correlated with u_i as well.
- *Bias direction:* hospitals that specialise in treating the most severe cases tend to look *artificially worse* on raw outcome comparisons – a downward bias on their apparent quality, even if their actual clinical care is excellent (the classic critique of unadjusted hospital “report cards”).

• **Example 4 (online platforms): product ratings and reviews.**

- *Model:* $\text{rating}_i = \beta_0 + \beta_1 \text{price}_i + u_i$, where rating_i is the average star rating left for product i .
- *What is in u_i :* the buyer’s idiosyncratic satisfaction with the specific unit received.
- *Why Assumption 1 fails:* only a small, non-random fraction of buyers leave a review, and the decision to review is itself correlated with the realised value of u_i – buyers with an extreme experience (very satisfied or very dissatisfied) are far more likely to post a review than buyers with an average experience. The observed sample of ratings is therefore selected on u_i .
- *Bias direction:* typically toward *polarisation* – observed rating distributions are more extreme (more 1-star and 5-star reviews, fewer 3-star) than the true population distribution of buyer satisfaction would be, and any estimated price–rating relationship inherits this distortion.

Technique note. In every example the diagnostic is the same: ask whether the event “unit i is in my sample” depends on u_i . If it does, Assumption 1 fails and OLS is biased in the direction implied by the sign of the induced correlation between the regressor and u_i .

Part (b): the measure of fit R^2 , and the ESS/TSS diagram

Definitions. Every fitted value decomposes the actual value into an explained part and a residual,

$$Y_i = \hat{Y}_i + \hat{u}_i, \quad i = 1, \dots, n,$$

and the course defines

$$\text{ESS} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \quad (\text{explained sum of squares}),$$

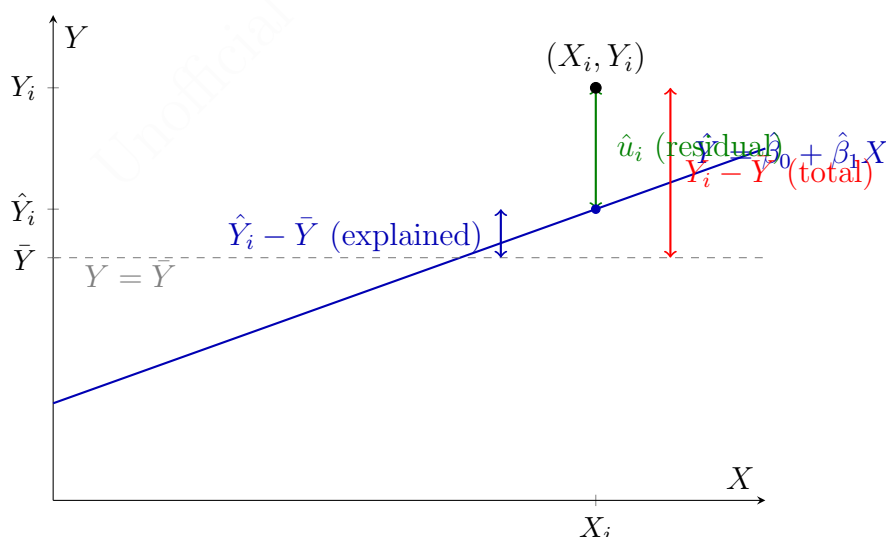
$$\text{TSS} = \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad (\text{total sum of squares}),$$

with the residual sum of squares $\text{SSR} = \sum_{i=1}^n \hat{u}_i^2$ satisfying the identity $\text{TSS} = \text{ESS} + \text{SSR}$ (proved in Question 2(b) of Question Set No. 5 by expanding $Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{u}_i$ and using $\sum_{i=1}^n \hat{u}_i = 0$, $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$). R^2 is then defined as the fraction of the sample variation in Y that is explained by X :

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}}.$$

Because $\text{SSR} \geq 0$ and (by the identity) $\text{TSS} \geq \text{ESS} \geq 0$, R^2 always lies in $[0, 1]$: $R^2 = 0$ when $\hat{\beta}_1 = 0$ (X explains nothing, $\hat{Y}_i = \bar{Y}$ for all i), and $R^2 = 1$ when every residual is zero (X explains everything, $Y_i = \hat{Y}_i$ for all i). A low R^2 (e.g. 0.051 for the Test Score/STR regression) is not itself “bad”: it simply says other, omitted factors drive most of the variation in Y .

Diagram: the ESS/TSS decomposition for a single observation. The official slide diagram could not be recovered from the source PDF (image-only content); it is redrawn below from its description in the Tut 4 transcript (“blue square = $\hat{Y}_i - \bar{Y}$ ”, “red square = $Y_i - \bar{Y}$ ”). For one observation (X_i, Y_i) , the vertical distance from the horizontal reference line $Y = \bar{Y}$ up to the data point is the *total* deviation $Y_i - \bar{Y}$; this splits into the *explained* deviation $\hat{Y}_i - \bar{Y}$ (from $\bar{Y}$ up to the fitted line) and the *residual* $\hat{u}_i = Y_i - \hat{Y}_i$ (from the fitted line up to the point).



The official remark: individual vs. aggregate comparison. The Tut 4 slides note that for *some particular* observation i , the squared total deviation $(Y_i - \bar{Y})^2$ can be smaller than the squared explained deviation $(\hat{Y}_i - \bar{Y})^2$ – yet summed across all n observations we always have $\text{TSS} \geq \text{ESS}$. This is not a contradiction: the identity $\text{TSS} = \text{ESS} + \text{SSR}$ holds term-by-term only after adding the cross term $2 \sum_{i=1}^n \hat{u}_i (\hat{Y}_i - \bar{Y})$, which vanishes only in

the *sum* (because $\sum_{i=1}^n \hat{u}_i = 0$ and $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$), not observation by observation. So a single point can have its fitted deviation overshoot its total deviation, while in aggregate the residual sum of squares $\text{SSR} \geq 0$ always pulls TSS back up to at least ESS.

Beyond the slides: a numerical mini-example. Take $n = 4$ with $X = (1, 2, 3, 4)$, $Y = (2, 3, 3, 8)$. Then $\bar{X} = 2.5$, $\bar{Y} = 4$, and the OLS fit is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = 1.8, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = -0.5,$$

giving fitted values $\hat{Y} = (1.3, 3.1, 4.9, 6.7)$. The per-observation squared deviations are:

i	X_i	Y_i	$(Y_i - \bar{Y})^2$	$(\hat{Y}_i - \bar{Y})^2$
1	1	2	4.00	7.29
2	2	3	1.00	0.81
3	3	3	1.00	0.81
4	4	8	16.00	7.29

At $i = 1$, $(Y_1 - \bar{Y})^2 = 4.00 < (\hat{Y}_1 - \bar{Y})^2 = 7.29$ – exactly the individual-point exception the slides warn about. Yet summing every column gives $\text{TSS} = 22.00$ and $\text{ESS} = 16.20$, so $\text{TSS} \geq \text{ESS}$ holds in aggregate as claimed, with $\text{SSR} = 22.00 - 16.20 = 5.80$ and

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = \frac{16.20}{22.00} = 0.7364.$$

Interpretation. About 73.6% of the sample variation in Y is explained by the linear fit on X in this toy example; the remaining 26.4% is left in the residuals.

Question 2 (Predicting weight from height)

Problem

A random sample of 100 20-year-old men is selected from a population and these men's height and weight are recorded. An OLS regression of weight on height yields

$$\widehat{\text{Weight}} = -79.24 + 4.16 \times \text{Height}, \quad (1)$$

where Weight is measured in pounds and Height is measured in inches.

- What is the regression's weight prediction for someone who is 64 inches tall? 68 inches tall? 72 inches tall?
- A man has a late growth spurt and grows 2 inches over the course of a year. What is the regression's prediction for the increase in this man's weight?

(Tutorial Question Set No. 4, Q2. No official solution is given on the Tutorial 4 slides – the “Question 2” slide is blank – so this solution is supplied in full.)

Solution

Part (a): predicted weight at three heights

Method 1: direct substitution into the fitted equation. Substitute each value of Height into $\widehat{\text{Weight}} = -79.24 + 4.16 \times \text{Height}$:

$$\text{Height} = 64 : \widehat{\text{Weight}} = -79.24 + 4.16(64) = -79.24 + 266.24 = 187.00 \text{ lbs,}$$

$$\text{Height} = 68 : \widehat{\text{Weight}} = -79.24 + 4.16(68) = -79.24 + 282.88 = 203.64 \text{ lbs,}$$

$$\text{Height} = 72 : \widehat{\text{Weight}} = -79.24 + 4.16(72) = -79.24 + 299.52 = 220.28 \text{ lbs.}$$

$\begin{aligned} \widehat{\text{Weight}}(64) &= 187.00, \\ \widehat{\text{Weight}}(68) &= 203.64, \\ \widehat{\text{Weight}}(72) &= 220.28 \text{ pounds.} \end{aligned}$

(Verified numerically with Python: $-79.24 + 4.16 \cdot h$ at $h = 64, 68, 72$ returns 187.0, 203.64, 220.28.)

Part (b): predicted weight gain from a 2-inch growth spurt

Method 1: direct substitution and subtraction. Pick any starting height H and evaluate the fitted line at H and $H + 2$; e.g. $H = 64$:

$$\widehat{\text{Weight}}(66) = -79.24 + 4.16(66) = 195.32, \quad \widehat{\text{Weight}}(64) = 187.00,$$

$$\Delta \widehat{\text{Weight}} = 195.32 - 187.00 = 8.32 \text{ lbs.}$$

Method 2: via the slope coefficient, $\hat{\beta}_1 \Delta X$. Because the fitted relationship is linear, the predicted change in $\widehat{\text{Weight}}$ for any change ΔHeight is simply the slope times the change – the intercept and the starting height cancel:

$$\Delta \widehat{\text{Weight}} = \widehat{\text{Weight}}(H + \Delta H) - \widehat{\text{Weight}}(H) = \hat{\beta}_1 \cdot \Delta H = 4.16 \times 2 = 8.32 \text{ lbs},$$

for *any* starting height H (this matches the difference computed by direct substitution in Method 1, and confirms the answer does not depend on which starting height was picked).

$$\Delta \widehat{\text{Weight}} = \hat{\beta}_1 \times 2 = 8.32 \text{ pounds.}$$

Interpretation. The slope $\hat{\beta}_1 = 4.16$ says that, comparing two 20-year-old men in this cross-sectional sample who differ in height by one inch, the taller man’s predicted weight is on average 4.16 lbs higher. The intercept $\hat{\beta}_0 = -79.24$ has no sensible standalone interpretation (it would predict negative weight at Height= 0), since Height= 0 is far outside the range of the data – a reminder that OLS fitted lines should not be extrapolated outside the observed range of X .

Beyond the slides: a cross-sectional caveat. Equation (1) is estimated from a single cross-section of 100 *different* men measured once; $\hat{\beta}_1$ captures the *association between height and weight across different people at a point in time*, not the effect of one particular man growing taller *over time*. Applying $\hat{\beta}_1 \times 2$ to “a man who grows 2 inches over a year” implicitly treats a within-person, longitudinal change as though it were the same as a between-person, cross-sectional comparison – i.e. it assumes the height–weight relationship that describes differences between already-grown men in the sample also describes how one man’s weight changes as he grows during a late growth spurt. This is an extrapolation of a cross-sectional association to a different (dynamic, individual-growth) setting, and is not guaranteed to hold: a growth spurt in a 20-year-old is typically accompanied by puberty-related increases in bone and muscle mass that need not track the population-average height–weight slope estimated from men who have already finished growing. The number 8.32 lbs is still the regression’s best available point prediction, but it should be reported together with this caveat rather than treated as a causal, individual-level growth forecast.

Question 3 (Consequences of $\hat{\beta}_1 = 0$ for $\hat{Y}_i$ and R^2)

Problem

- (a) Show that the average of the OLS predicted values $\hat{Y}_i$, $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i$, is equal to the sample average of Y , $\bar{Y}$.
- (b) Show that if the OLS estimator $\hat{\beta}_1 = 0$, then $\hat{Y}_i = \hat{\beta}_0 = \bar{Y}$.
- (c) Show that if (b) holds, then $R^2 = 0$.

(Tutorial Question Set No. 4, Q3.)

Solution

Part (a): $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y}$

Method 1: via $Y_i = \hat{Y}_i + \hat{u}_i$ and $\sum_{i=1}^n \hat{u}_i = 0$ (official route). By definition of the residual, $Y_i = \hat{Y}_i + \hat{u}_i$ for every i . Summing over $i = 1, \dots, n$,

$$\sum_{i=1}^n Y_i = \sum_{i=1}^n \hat{Y}_i + \sum_{i=1}^n \hat{u}_i.$$

The FOC of the OLS minimisation problem gives $\sum_{i=1}^n \hat{u}_i = 0$ (Question 2(a) of Question Set No. 3), so $\sum_{i=1}^n Y_i = \sum_{i=1}^n \hat{Y}_i$. Dividing both sides by n ,

$$\boxed{\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y}.}$$

Method 2: via $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ (substitute the closed-form intercept). Since $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$,

$$\sum_{i=1}^n \hat{Y}_i = \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i) = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n X_i = n\hat{\beta}_0 + n\hat{\beta}_1 \bar{X} = n(\hat{\beta}_0 + \hat{\beta}_1 \bar{X}).$$

Using the OLS formula $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$,

$$\sum_{i=1}^n \hat{Y}_i = n[(\bar{Y} - \hat{\beta}_1 \bar{X}) + \hat{\beta}_1 \bar{X}] = n\bar{Y} \implies \frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y},$$

which reproduces the official second solution and matches Method 1 without needing the residual identity at all.

Part (b): if $\hat{\beta}_1 = 0$ then $\hat{Y}_i = \hat{\beta}_0 = \bar{Y}$ for all i

Method 1: direct substitution. If $\hat{\beta}_1 = 0$, the fitted value at every i reduces to

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i = \hat{\beta}_0 + 0 \cdot X_i = \hat{\beta}_0.$$

By the OLS formula, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = \bar{Y} - 0 \cdot \bar{X} = \bar{Y}$. Combining the two equalities,

$$\hat{Y}_i = \hat{\beta}_0 = \bar{Y} \quad \text{for every } i = 1, \dots, n.$$

In words: if X has an estimated slope of exactly zero, the fitted line degenerates into the horizontal line at the sample mean of Y – X contributes nothing to the prediction of any individual Y_i .

Part (c): if (b) holds, then $R^2 = 0$

Method 1: directly from $ESS = 0$ (official Solution 1). Since $\hat{Y}_i = \bar{Y}$ for every i (part (b)), each term of the explained sum of squares is zero:

$$ESS = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \sum_{i=1}^n (\bar{Y} - \bar{Y})^2 = \sum_{i=1}^n 0 = 0.$$

Hence

$$R^2 = \frac{ESS}{TSS} = \frac{0}{TSS} = 0 \quad (\text{provided } TSS > 0).$$

Method 2: via the residuals, $SSR = TSS$ (official Solution 2). Alternatively, if $\hat{\beta}_1 = 0$ the residual at every i is

$$\hat{u}_i = Y_i - \hat{Y}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i = Y_i - \hat{\beta}_0 = Y_i - \bar{Y},$$

so

$$SSR = \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2 = TSS.$$

Since $TSS = ESS + SSR$ always holds, $SSR = TSS$ forces $ESS = 0$, and again $R^2 = ESS/TSS = 0$. This route reaches the same conclusion through the residuals rather than through the fitted values directly – a useful cross-check whenever computing ESS directly is awkward.

Beyond the slides: the edge case $TSS = 0$. The division $R^2 = ESS/TSS$ is only defined for $TSS > 0$. $TSS = 0$ happens exactly when $Y_i = \bar{Y}$ for every i – i.e. Y is literally constant across the sample, so there is zero sample variation to explain in the first place. In that degenerate case $R^2 = 0/0$ is undefined rather than equal to 0; the course (and this solution) implicitly assumes $TSS > 0$, which holds for any sample with genuine variation in Y .

Question 4 (CAPM: comparing assets with equal β but different α)

Problem

As explained in class, we can use OLS to estimate the Capital Asset Pricing Model (CAPM). The linear regression model is defined as

$$(R - R_f) = \alpha + \beta(R_m - R_f) + u, \quad (2)$$

where $R - R_f$ denotes the excess return of the stock (or hedge fund), $R_m - R_f$ denotes the excess return of the market (e.g., R_m can be the return of S&P 500), and R_f denotes the return of the risk-free asset (e.g., US government bond). In practice, we report the OLS estimates $\hat{\alpha}$ and $\hat{\beta}$ for the stock (or fund). Under the standard CAPM theory, $\alpha = 0$. Therefore, if we find $\hat{\alpha}$ is significantly positive for a certain stock (or fund), then it is especially valuable as it can beat the prediction from CAPM (i.e., it has a premium that cannot be explained by CAPM). Compare the return and risk of a stock (or a fund) with different values of α but the same value of β . What do you find? [Hint: find its expected value and variance.] (*Tutorial Question Set No. 4, Q4. No official solution is given on the Tutorial 4 slides – both “Question 4” slides in the deck are blank – so this solution is supplied in full, extending the Lecture 4 CAPM material.*)

Solution

Setup. Write $Y = R - R_f$ (excess asset return) and $X = R_m - R_f$ (excess market return), so the model is $Y = \alpha + \beta X + u$ with the Least Squares Assumptions holding, in particular Assumption 1, $E[u | X] = 0$, which by the Law of Iterated Expectations implies $E[u] = 0$ and, since $\text{Cov}(X, u) = E[Xu] - E[X]E[u] = E[XE[u | X]] - 0 = 0$, also $\text{Cov}(X, u) = 0$.

Method 1: expected value and variance (the route suggested by the hint)

Expected excess return. Taking expectations of the model and using $E[u] = 0$,

$$E[R - R_f] = E[\alpha + \beta(R_m - R_f) + u] = \alpha + \beta E[R_m - R_f].$$

Consider two assets A and B with the *same* β but $\alpha_A > \alpha_B$. Subtracting,

$$E[R_A - R_f] - E[R_B - R_f] = \alpha_A - \alpha_B > 0,$$

so the asset with the strictly higher α has a strictly higher expected excess return, for *any* common value of $E[R_m - R_f]$.

Variance (risk). Because α and β are constants,

$$\text{Var}(R - R_f) = \text{Var}(\alpha + \beta(R_m - R_f) + u) = \beta^2 \text{Var}(R_m - R_f) + \text{Var}(u) + 2\beta \text{Cov}(R_m - R_f, u).$$

Assumption 1 gives $\text{Cov}(R_m - R_f, u) = 0$, so the cross term vanishes and

$$\boxed{\text{Var}(R - R_f) = \beta^2 \text{Var}(R_m - R_f) + \text{Var}(u),}$$

which does *not* involve α at all: α is a pure location shift of the excess-return distribution and has no effect on its dispersion. Risk here has two components – *systematic* risk $\beta^2 \text{Var}(R_m - R_f)$, driven by exposure to the market, and *idiosyncratic* risk $\text{Var}(u)$, specific to the asset. Holding β fixed makes the systematic component identical across the two assets; note that $\text{Var}(u)$ is asset-specific and could in principle still differ between A and B even with equal β – but any such difference is unrelated to, and not driven by, α .

Conclusion. Holding β (and hence systematic risk) and $\text{Var}(u)$ fixed, the asset with the higher α offers a strictly higher expected excess return *at the same level of risk*. This is exactly why a statistically significant, positive $\hat{\alpha}$ is prized in practice: it signals an abnormal return relative to what CAPM predicts for that level of market risk – performance that cannot be explained simply by the asset being riskier. Under the CAPM null ($\alpha = 0$ for every asset), no asset can systematically offer a higher expected excess return without also taking on more systematic risk (a higher β); a nonzero, significant $\hat{\alpha}$ is evidence against the CAPM holding exactly for that asset.

Method 2: framing $\hat{\alpha}$ as a hypothesis test (a genuinely different lens)

Method 1 compares the two assets *structurally*, through their population moments. In an empirical application we instead test whether an estimated $\hat{\alpha}$ is different from the CAPM benchmark of zero, using the same four-beat layout as the CAPM/AAPL beta test in Lecture 6.

Hypotheses. $H_0 : \alpha = 0$ (CAPM holds exactly for this asset) vs. $H_1 : \alpha \neq 0$ (or the one-sided $H_1 : \alpha > 0$ if the question is specifically about “beating” CAPM).

Test statistic. $t^{\text{act}} = \frac{\hat{\alpha} - 0}{\text{SE}(\hat{\alpha})}$.

Decision rule. Reject H_0 at the 5% level iff $|t^{\text{act}}| > 1.96$ (two-sided), or $t^{\text{act}} > 1.645$ for the one-sided alternative that the fund beats CAPM.

Conclusion in context. A large, positive t^{act} (equivalently, a small p -value) means the estimated premium $\hat{\alpha}$ is too large to be plausibly explained by sampling noise around $\alpha = 0$; combined with Method 1’s finding that α does not affect $\text{Var}(R - R_f)$, this is read as *genuine outperformance at no extra risk*, not merely a riskier bet that happened to pay off.

Beyond the slides: a numerical illustration. Suppose the market has expected monthly excess return $E[R_m - R_f] = 0.6\%$ with standard deviation 4%, and both funds share $\beta = 1.2$ and idiosyncratic risk $\text{sd}(u) = 2\%$ (so $\text{Var}(u) = 0.02^2 = 0.0004$). Fund A has $\alpha_A = 0$ (CAPM holds exactly); Fund B has $\alpha_B = 0.3\%$ per month. Then for both funds,

$$\begin{aligned}\text{Var}(R - R_f) &= (1.2)^2(0.04)^2 + 0.0004 = 0.002304 + 0.0004 = 0.002704, \\ \text{sd}(R - R_f) &= \sqrt{0.002704} = 5.20\%,\end{aligned}$$

identical for A and B since β and $\text{Var}(u)$ are unchanged. But the expected excess returns differ:

$$E[R_A - R_f] = 0 + 1.2(0.6\%) = 0.72\%, \quad E[R_B - R_f] = 0.3\% + 1.2(0.6\%) = 1.02\%.$$

Fund B earns 0.30 pp more per month, at the identical risk $\text{sd} = 5.20\%$.

Fund B 's $\alpha = 0.3\%$ is a pure “free lunch” relative to Fund A under CAPM: same market exposure, same total volatility, strictly higher expected return.

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Method Map

Q.	Core topic	Methods included
1	Selection bias; R^2 , ESS, TSS	(a) diagnostic: does sample membership depend on u_i ? – official investment-gain example plus mutual-fund, hospital, and online-review examples. (b) $R^2 = \text{ESS}/\text{TSS}$ definition; pgfplots redraw of the official ESS/TSS diagram; numerical mini-example reconciling the individual-point exception with $\text{TSS} \geq \text{ESS}$ in aggregate.
2	Predicting weight from height	Direct substitution into the fitted line; slope-times- ΔX shortcut for a marginal change; cross-sectional-vs-longitudinal extrapolation caveat.
3	Consequences of $\hat{\beta}_1 = 0$	FOC/residual-sum route and closed-form-intercept route for $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y}$; direct substitution for $\hat{Y}_i = \hat{\beta}_0 = \bar{Y}$; ESS = 0 route and SSR = TSS route for $R^2 = 0$; TSS = 0 edge case.
4	CAPM: same β , different α	Moment-based route ($E[\cdot]$, $\text{Var}(\cdot)$ under Assumption 1); hypothesis-test framing of $\hat{\alpha}$ (four-beat layout); numerical illustration of the “free lunch.”

Technique Summary

- Selection bias is a sample-membership problem.** Assumption 1 fails whenever whether unit i is observed at all (or observed with a particular value of X_i) depends on u_i ; the direction of the resulting bias follows the sign of the induced correlation between X and u .
- R^2 is a ratio of sums of squares, not a per-observation statement.** $\text{TSS} = \text{ESS} + \text{SSR}$ holds only after summing across all i (the cross term $\sum_{i=1}^n \hat{u}_i(\hat{Y}_i - \bar{Y})$ vanishes in aggregate via $\sum_{i=1}^n \hat{u}_i = 0$, $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$), so individual observations can violate the “pattern” that the aggregate identity suggests.
- Linear predictions travel in slopes.** For a linear fit $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, the predicted change for any ΔX is always $\hat{\beta}_1 \Delta X$, regardless of the starting value of X – but a cross-sectional slope answers a between-units comparison, not a within-unit, over-time question.
- $\hat{\beta}_1 = 0$ collapses the regression to the sample mean.** When the slope is exactly zero, $\hat{Y}_i = \bar{Y}$ for every i , ESS = 0, and $R^2 = 0$: X is contributing nothing beyond what $\bar{Y}$ alone would say.
- In CAPM, α shifts location but not spread.** Under Assumption 1, $\text{Cov}(R_m - R_f, u) = 0$, so $\text{Var}(R - R_f) = \beta^2 \text{Var}(R_m - R_f) + \text{Var}(u)$ is unaffected by α ; comparing same- β assets isolates α as a pure expected-return premium at unchanged risk.

Final Answers

- 1(a)** Investment-gain/experience (official); mutual-fund databases (survivorship of successful funds); hospital outcome rankings (severity-driven selection); online reviews (polarised self-selection into reviewing).
- 1(b)** $R^2 = \text{ESS}/\text{TSS} = 1 - \text{SSR}/\text{TSS} \in [0, 1]$; diagram above; numerical example gives $\text{TSS} = 22.00$, $\text{ESS} = 16.20$, $\text{SSR} = 5.80$, $R^2 = 0.7364$, with an individual point ($i = 1$) where $(Y_1 - \bar{Y})^2 = 4.00 < (\hat{Y}_1 - \bar{Y})^2 = 7.29$ despite $\text{TSS} \geq \text{ESS}$ overall.
- 2(a)** $\widehat{\text{Weight}}(64) = 187.00$, $\widehat{\text{Weight}}(68) = 203.64$, $\widehat{\text{Weight}}(72) = 220.28$ pounds.
- 2(b)** Predicted increase $= \hat{\beta}_1 \times 2 = 8.32$ pounds (with the caveat that this extrapolates a cross-sectional association to a within-person, over-time change).
- 3(a)** $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i = \bar{Y}$.
- 3(b)** $\hat{Y}_i = \hat{\beta}_0 = \bar{Y}$ for every i , when $\hat{\beta}_1 = 0$.
- 3(c)** $\text{ESS} = 0 \Rightarrow R^2 = 0$ (assuming $\text{TSS} > 0$; R^2 is undefined if $\text{TSS} = 0$).
- 4** $E[R - R_f] = \alpha + \beta E[R_m - R_f]$ (higher α raises expected excess return); $\text{Var}(R - R_f) = \beta^2 \text{Var}(R_m - R_f) + \text{Var}(u)$ (independent of α). At equal β , a higher- α asset has strictly higher expected excess return at unchanged risk – illustrated numerically by $E[R_A - R_f] = 0.72\%$ vs. $E[R_B - R_f] = 1.02\%$ with identical $\text{sd} = 5.20\%$.